# Hazard Analysis
# Software Engineering

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Table 1: Revision History

| **Date** | **Developer(s)** | **Change** |
|---|---|---|
| October 15, 2026 | Owen Johnson | Introduction, Scope and Purpose Updated |
| October 17, 2026 | Humna Saeed | Section 3: System Boundaries & Components Draft |
| October 17, 2026 | Andreas Sideris | AI Disclosure Updated |
| October 18, 2026 | Andreas Sideris | FMEA Table Completed |
| October 18, 2026 | Andreas Sideris | Reflection Added |
| October 19, 2026 | Owen Johnson | Reflection Added |

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# 1 Introduction

This document is a hazard analysis of the RealmForge system, which is a roguelite video game. A roguelite game is defined by different attempts to clear the game being varied from each other in some way, and being mostly self-contained with a small amount of certain progression carrying over, with the goal that each attempt feels unique. RealmForge intends to provide an engaging game experience with lots of variability while collecting data from users about the effect of this variability to further enhance the game experience through intuitive usability, fun game design/player experience, and replayability.

Using Nancy Leveson's work (referenced in lecture), a hazard is defined as a property/condition in the system that, when combined with an environment condition, has potential to cause harm or damage. For RealmForge's purposes, **harm** can be described as:

- Player dissatisfaction
- Performance issues
- Poor usability testing data

# 2 Scope and Purpose of Hazard Analysis

The scope of this document is to provide an analysis of the system as it pertains to potential hazards. The analysis will identify hazards present in the components/boundaries of the system, provide mitigation strategies for each hazard, and provide the resulting safety and security requirements.

This analysis is important, as certain losses can be incurred by the hazards if left unchecked, such as:

- Lower player count and interest
- Player frustration
- Unintended behavior, such as bugs, lag, or crashes
- Lack of usability testing data to work from

# 3 System Boundaries and Components

The following section identifies the system boundaries of RealmForge and the game's major components. These components help identify and evaluate potential hazards that could affect gameplay, player progress, and the overall functionality of the game.

### System Boundaries

RealmForge is a locally running game that allows players to configure game properties, explore different areas, participate in combat encounters, and purchase upgrades using resources earned throughout gameplay. The system is responsible for processing player inputs, displaying gameplay information, updating the game based on player actions, and saving player progress locally.

External components include the user's operating system, input devices such as a keyboard, mouse, or supported controller, and output devices such as a monitor and speakers or headphones.

### System Components

#### Property Configuration

Property configuration allows for the selection and application of game property variations prior to starting a game session. These selections vary based on player preference and influence the overall gameplay experience.

#### World Navigation

This component handles player movement as they navigate the game world, allowing players to explore different locations as they progress through the game.

#### Combat System

The combat system is responsible for handling combat encounters between the player and opponents within the game. This includes player attack actions, interactions, and the outcomes of combat encounters.

#### Progression and Rewards

This component is responsible for tracking resources earned throughout gameplay, as well as player progression and rewards received from completing encounters.

#### Central Hub

The central hub allows players to purchase items and upgrades using resources earned throughout gameplay. Players return to the hub when starting a new game run, allowing them to prepare for their next session.

#### Progress Management

The progress management system is responsible for saving player progress locally to ensure that player progress is preserved for future sessions.

#### User Interface

The user interface component is responsible for displaying gameplay information and allowing player interactions. This includes interactions with menus, combat options, world navigation, the central hub, and other game features.

## 4 Critical Assumptions

- The application can be correctly installed and launched on a machine running a Windows Operating System (Windows 10 or 11). The application can be closed without causing fatal system crashes in the hardware used.

- The application will not maliciously introduce any computer viruses or malware to the user's machine.
- The application will not cause crashes or hardware meltdowns to input periphals (e.g., keyboard and mouse).
- The contents of the software (in the RealmForge game) will not contain any inappropriate language, or extreme mature themes/images.
- The application will not play any loud audio cues or sounds that can cause damage to the user's hearing (at or exceeding 85dB).

# 5 Failure Mode and Effect Analysis

Table 2: Failure Mode and Effect Analysis (FMEA)

| **Component** | **Failure Mode** | **Effects of Failure** | **Causes of Failure** | **Recommended Action** | **Safety Requirements** | **Ref.** |
|---|---|---|---|---|---|---|
| Property Configuration | System is unable to recognize the user's connected input device (such as a controller), when selected in the feature variation menu | Users waste their time and can become confused/frustrated when attempting to troubleshoot input device connection | A. System does not provide sufficient guidance to the user to connect their input device<br>B. System does not correctly monitor device interfaces to receive input (Bluetooth, USB) | A. Provide detailed guidance to the user on how to connect each input device; prompt the user to select the device they wish to connect (PlayStation DualSense controller, Xbox Series X controller, etc.) and provide guidance specific to that device<br>B. Monitor both USB and Bluetooth interfaces for connection to each type of supported input device when the user opts to connect a new device | CREATE NEW REQS FOR THIS | H1-A |

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Table 2 – Continued

| **Component** | **Failure Mode** | **Effects of Failure** | **Causes of Failure** | **Recommended Action** | **Safety Requirements** | **Ref.** |
|---|---|---|---|---|---|---|
| World Navigation | User is unable to understand where their player character can move next in the overall game world | User can become confused/frustrated, and waste gameplay time staying in the main game world instead of progressing forwards | System does not provide adequate feedback to the user regarding where the player character can move next; valid locations are indistinguishable from invalid locations, or a possible path to take to reach the end of the game is unclear | Highlight valid locations that the player character can move to in the game world, and indicate invalid locations. Provide further guidance to the user regarding a possible path forward if an extended period of time is spent attempting to reach invalid locations | CREATE NEW REQS FOR THIS | H2-A |
| Combat System | System displays flashing or intense graphics | Users with photosensitive epilepsy can potentially suffer an epileptic seizure | The system displays excessive flashing graphics without warning users with photosensitive epilepsy | Include a detailed flashing lights or epilepsy warning when the system launches, and ensure the game does not display excessive flashing graphics | CREATE NEW REQS FOR THIS | H3-A |

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Table 2 – Continued

| **Component** | **Failure Mode** | **Effects of Failure** | **Causes of Failure** | **Recommended Action** | **Safety Requirements** | **Ref.** |
|---|---|---|---|---|---|---|
| Progression and Rewards | Rewards provided to the user for succeeding in an enemy encounter are lost when exiting the encounter to the game world | Gameplay progression is interrupted by earned rewards not being given to the user, preventing them from using rewards to improve their in-game capabilities; this can cause the user to become frustrated | Variables storing enemy encounter rewards are erroneously overwritten by another value during transitions between game scenes | Retain and update enemy encounter reward variables (during encounters) in an additional memory location separate from the combat system, to ensure a backup value is available | CREATE NEW REQS FOR THIS | H4-A |

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Table 2 – Continued

| **Component** | **Failure Mode** | **Effects of Failure** | **Causes of Failure** | **Recommended Action** | **Safety Requirements** | **Ref.** |
|---|---|---|---|---|---|---|
| Central Hub | System does not return game state and display to the Central Hub menu following a failed game run (user is defeated) | User is unable to continue the correct flow of gameplay (returning to the Central Hub menu, starting again), and may need to restart the software to resolve the issue, causing frustration | Software errors/faults prevent the system from correctly transitioning between game scenes (end of game run/user is defeated to Central Hub menu) | Extensively test and verify major game state transitions (user defeat to Central Hub menu, reaching a game world area to an enemy encounter) for correctness. Game state transitions can be verified against a state machine for clarity | N/A | H5-A |
| Progress Management | System application unexpectedly crashes or closes completely due to an error | Most recent user save data is corrupted and unrecoverable; the user can be frustrated due to lost progress | System does not store any backup user save data | Regularly store user save data in multiple locations; revert to an intact recent saved state on system failure | CREATE NEW REQS FOR THIS | H6-A |

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Table 2 – Continued

| **Component** | **Failure Mode** | **Effects of Failure** | **Causes of Failure** | **Recommended Action** | **Safety Requirements** | **Ref.** |
|---|---|---|---|---|---|---|
| Progress Management | System attempts to initialize the game using corrupted save data | The system is unable to correctly initialize the game and enable gameplay; the user can be confused on how to proceed using their original saved game data, and can become frustrated | System does not take steps to validate user save data | Validate user save data against previous snapshots; if save data is corrupted, revert to an intact recent saved state | CREATE NEW REQS FOR THIS | H6-B |



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Table 2 – Continued

| **Component** | **Failure Mode** | **Effects of Failure** | **Causes of Failure** | **Recommended Action** | **Safety Require-ments** | **Ref.** |
|---|---|---|---|---|---|---|
| User Interface | System displays game and interface graphics that are difficult for colourblind users to interpret | Colourblind users are unable to progress efficiently in the game, causing frustration; the game is not accessible to as many users as possible | Displayed symbols in game and interface graphics are difficult to distinguish for colourblind users (colours such as red and green are used as a primary method of differentiation), or colourblind-accessible graphics filters are not provided | Avoid relying only on certain colours to convey interface and game information (red and green), and provide appropriate colourblind-accessible graphics filter settings to the user | CREATE NEW REQS FOR THIS | H7-A |

## 6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

## 7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

# 8 Appendix — Generative AI Use Disclosure

## AI Tools Used

- OpenAI GPT-6
- Google Search AI Overview (Google Gemini)

## How and Where Used

AI tools were used for assistance with LaTeX syntax and commands, during the development of the FMEA table section, and when developing LaTeX formatting for the document as a whole. Specifically, AI tools were used for assistance in creating the LaTeX table used for the FMEA summary and creating the list of tables, as well as resolving various minor formatting bugs. AI tools were also used to create the LaTeX custom requirement format macro used in the document, that was also used in the SRS. In this section (FMEA), AI tools were not used to generate any text or wholly generate any of the ideas or conclusions presented.

## Verification of AI-Generated Content

Any LaTeX syntax or code provided by AI tools was verified by testing compilation in LaTeX documents, and ensuring the resulting output was correctly formatted and matched the needs of the document. No ideas or text were newly generated by AI tools and presented in the document.

# 9 Appendix — Reflection

**Reflection Questions Answered:**

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

**Andreas:**

1. When writing this deliverable, our team worked well to synchronize our decisions between sections. While I primarily focused on the FMEA table, I was able to discuss decisions my teammates had made on other vital sections (such as System Boundaries and Components) with them, clarify uncertainties, and use the output of our discussions to guide my work.
2. One pain point I experienced during this deliverable was finding suitable failure modes and hazards for our software, since it is intended to be a video game (it does not have many obvious physical safety risks). This was resolved by considering other ways failure of the system could result in harm, such as it being inaccessible to users with photosensitive epilepsy, or colourblind users. It was also found that other failures of the system could result in harm (in the expanded sense used in this project), by frustrating users and degrading the user experience overall.
3. Some risks had been considered before, such as failures of the system resulting in corrupted user save data, or unexpected behavior (that could frustrate the user), but specific risks concerning colourblind users and users with photosensitive epilepsy were newly found and formalized. These risks were found by considering the ways in which failures of the system could cause harm to specific user groups, if they were not accounted for in the design (as these risks could be missed without careful consideration). This led to finding a physical safety hazard (the potential for epileptic seizures in those with photosensitive epilepsy), as well as a potential risk of preventing colourblind users from playing the game as others can.
4. One risk can involve the risk of preventing certain users from being able to use a software product (as discussed above, in regards to accessibility). This risk is important to consider because it can result in real harm; users

with specific conditions can feel alienated when they aren't able to use or enjoy software products that others do. Another risk involves the risk of users becoming confused or frustrated when using the software product (as discussed in the FMEA analysis, when users' saved progress is lost, or the user interface is difficult to understand). This must be considered, as if most users become frustrated when using a software product, it will fail commercially/publicly in favour of other products that are designed with the user in mind.

**Humna:**

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**Christopher:**

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**Andy:**

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**Owen:**

1. When writing this deliverable, I think the team communicated well when working on different sections. The team asked for help when needed, and noticed when certain portions of divided work relied on other parts. The intro, scope and purpose were all intertwined with the other sections, with discussion taking place to ensure things fit together.

2. Some pain points that were experienced during this deliverable were time management, and finding adequate content to discuss in a relatively low risk, non safety critical system. Since the project is a video game it was harder to find physical risks, and this was resolved by including hazards/risks that are not immediately obvious, such as colourblind accessibility and interactions with photosensitive epilepsy. The pain point of time management was mainly because of the deliverable going through reading week, where schedules are uncertain and much different from a typical school week. This was resolved by an increase in Microsoft Teams activity, ensuring everyone was in the loop.

3. Some risks were thought of before this document, including loss of save data, game crashes, and other unintended behavior. These were all immediately identifiable risks that came about when initial discussions of developing a game were happening. Other, more specific risks came about later (during this document), such as users with photosensitive epilepsy having seizures due to flashing lights (this was discussed as a possibility in lecture). Specific risks such as these required extra thought about the design of the game, and not just the implementation.

4. 2 types of risk (other than physical) for software products include:

    - Barriers to Entry - Various design choices can cause barriers to entry for some users. For example, if similar colours are used and intended to be differentiated, this can cause colourblind users not to be able to interact with the game as intended.
    - Unsatisfied Users - Many design choices affect users' enjoyment of the game. Increased frustration/confusion in figuring out how to interact with the game can cause users to be unsatisfied with their experience, which can lead to consequences like negative game reputation/reviews or game refunds for paid games.